
Baseline Controls Temper Novel-Mechanism Claims from Sparse Protein Language Model Features

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Abstract

Protein language models encode residue-level regularities that support structure, function, and mutation-effect prediction, but it remains unclear when internal features should be treated as new biological hypotheses rather than rediscoveries of known annotations or artifacts of high-dimensional search. We examine this question in a compact, reproducible workflow that combines sparse feature association, motif exclusion, baseline controls, and internal intervention. We run ESM-2-8M with pretrained layer-4 INTERPLM sparse autoencoder features on 16 short PROTEINGYM deep mutational scanning assays, covering 12,085 single substitutions and 688 residue positions. The strongest sparse feature per assay correlates with residue-level mutation sensitivity, with median absolute Spearman correlation 0.547. However, this does not significantly exceed raw ESM-2 hidden dimensions (median 0.544; one-sided Wilcoxon $p = 0.798$) or shuffled sparse-feature controls (median shuffled 95th percentile 0.543; $p = 0.217$). We identify three low ELM/PROSITE alignment feature-assay candidates. One exploratory intervention, ablating INTERPLM feature 6419 in YNZC_BACSU_Tsuboyama_2023_2JVD, changes ESM-2 masked-marginal variant scores more at high-activation positions than at low-activation positions (effect 0.0223 log-prob units; Mann-Whitney $p = 0.00495$). These results do not establish a novel biological mechanism. They instead support a cautious triage protocol for turning protein-model features into testable hypotheses.

1 Introduction

Protein language models now score variants, guide protein engineering, and provide representations for structure and function prediction Rives et al. [2021], Lin et al. [2023], Notin et al. [2023]. Their success creates a scientific opportunity: if a model has learned a residue pattern that biologists have not yet annotated, that internal pattern could become a testable hypothesis. The hard part is separating that possibility from a weaker and more common result, where an internal feature merely correlates with known motifs, conservation, assay noise, or a favorable multiple-comparisons draw.

Our main question. We ask whether sparse features in a real protein language model identify functionally important protein positions that are not aligned with known motif annotations, and whether intervening on those features changes the model’s mutation-effect predictions. This question is narrower than general interpretability. It asks for evidence that a feature is functionally relevant, not already explained by implemented ELM/PROSITE motif scans, and involved in the computation used for masked-marginal scoring.

Recent work gives strong reasons to ask this question. INTERPLM decomposes ESM-2 embeddings into sparse features that recover binding sites, motifs, domains, and other curated biological concepts Simon and Zou [2025]. InterProt and related work argue that sparse autoencoder features can move from model interpretation toward biological hypothesis generation Adams et al. [2025]. Motif-focused sparse autoencoders further show that latent features can align with known short linear motifs and fitness landscapes Hou et al. [2025]. However, the presence of interpretable sparse

features is not enough to claim novelty. Short proteins and thousands of features make high residue-level correlations easy to find unless raw-coordinate and shuffled-position baselines are included.

We propose FEATURETRIAGE, a conservative pilot workflow for PLM-derived biological hypothesis triage. The workflow extracts INTERPLM layer-4 sparse features from ESM-2-8M, compares feature activations to PROTEINGYM residue-level DMS sensitivity, screens candidates against ELM and PROSITE motif scans on target sequences and a SWISS-PROT sample, and ablates selected decoder directions during ESM-2 masked-marginal scoring.

Quantitative preview. Sparse features frequently mark functionally sensitive positions: the median best absolute Spearman correlation between an SAE feature and DMS residue sensitivity is 0.547 across 16 assays. The conservative controls change the interpretation. The median best raw hidden dimension reaches 0.544, the median shuffled-feature 95th percentile is 0.543, and neither paired comparison supports an aggregate SAE advantage. We find three low-motif-alignment candidates, and one candidate intervention, feature 6419 in YNZC_BACSU_Tsuboyama_2023_2JVD, produces a high-minus-low activation effect of 0.0223 masked-score log-prob units.

Our contributions are:

- We propose a baseline-controlled workflow for triaging unannotated residue-level PLM features as biological hypotheses.
- We conduct a real ESM-2-8M and INTERPLM experiment on 16 PROTEINGYM assays with raw hidden-dimension, shuffled-position, and motif-alignment controls.
- We show that feature-function correlations alone are not sufficient evidence for novelty: sparse features do not significantly outperform the strongest raw or shuffled baselines in this pilot.
- We identify one exploratory intervention candidate, INTERPLM feature 6419, whose ablation changes masked-marginal variant scores preferentially at high-activation positions.

Section 2 reviews PLM interpretability and motif resources. Section 3 describes the workflow, controls, and intervention. Section 4 reports the aggregate, candidate, and ablation results. Section 5 interprets the evidence and limitations.

2 Related Work

Protein language models and mutation-effect benchmarks. Protein language models learn representations from large protein-sequence corpora and can support downstream biological prediction without direct supervision for each task. Earlier work showed that large protein sequence models encode biological structure and function Rives et al. [2021], and ESM-2-family models scale this idea to atomic-level structure prediction Lin et al. [2023]. PROTEINGYM standardizes deep mutational scanning and clinical variant benchmarks for evaluating protein fitness and design models Notin et al. [2023]. We use PROTEINGYM because it provides direct mutation-effect measurements, making it suitable for asking whether a residue-level internal feature marks positions where mutations are experimentally deleterious.

Sparse features in protein language models. Sparse autoencoders have become a central tool for decomposing dense model activations into higher-dimensional sparse features. INTERPLM trains sparse autoencoders on ESM-2 embeddings and shows that sparse features align with many curated concepts, while raw neurons show less conceptual alignment Simon and Zou [2025]. Gujral et al. [2025] similarly use sparse autoencoders and transcoders to extract interpretable protein-level and amino-acid-level features from PLM representations. Adams et al. [2025] frame this line of work as a path from mechanistic interpretability to biological hypothesis generation. We build on this view but add a stricter empirical standard: an unannotated feature must be compared to raw hidden dimensions, shuffled sparse controls, and motif scans before it is treated as a candidate.

Motif discovery and annotation controls. Motifs provide a natural residue-level bridge between model internals and known biology. The ELM resource catalogs eukaryotic short linear motifs Kumar et al. [2024], and PROSITE provides curated protein patterns, profiles, and rules for protein-domain and functional-site annotation Sigrist et al. [2013]. MotifAE adapts sparse autoencoders for motif discovery by encouraging local activation coherence and reports improved recovery of known ELM motifs Hou et al. [2025]. Our experiment uses ELM and PROSITE differently: not as the

prediction target, but as a conservative exclusion filter for claims of candidate novelty. We also scan a SWISS-PROT sample from UniProtKB The UniProt Consortium [2025] to estimate whether candidate feature activations align with motif-positive residues outside the target assays.

Why causal checks matter. Feature-label association is not enough to infer mechanism. A feature can correlate with DMS sensitivity because of conservation, local structure, motif-database incompleteness, or the large number of tested features. Interventions provide a stronger test by asking whether changing the feature changes model computation. Our ablations are still internal model interventions, not wet-lab validation, but they help separate features that merely track a label from features that participate in ESM-2 masked-marginal scoring.

3 Methodology

Task formulation. For each assay, we represent a target protein as a sequence of L residues. We aggregate single-mutant DMS measurements to a residue-level sensitivity value y_i , where larger values indicate that substitutions at position i are more deleterious on average. For a sparse feature f , we compute its wild-type activation $a_{i,f}$ at each residue position. The main association score is the within-assay Spearman correlation

$$\rho_f = \text{Spearman}((a_{1,f}, \dots, a_{L,f}), (y_1, \dots, y_L)). \quad (1)$$

We rank features by $|\rho_f|$ because a sparse feature can mark either protected high-sensitivity residues or permissive low-sensitivity residues.

Data. We use 16 short substitution assays from local PROTEINGYM v1 DMS files. The targets have length 37–52 residues and jointly contain 12,085 single substitutions over 688 residue positions. We also use 250 reviewed SWISS-PROT proteins sampled from the local UniProtKB/Swiss-Prot FASTA file for motif-alignment checks. Motif scans use 482 compiled patterns: 353 ELM patterns and 129 PROSITE patterns parsable by the implemented converter.

Model and sparse features. We run facebook/esm2_t6_8M_UR50D through Hugging Face Transformers and extract layer-4 hidden states. We then apply the pretrained Elna/InterPLM-esm2-8m layer-4 sparse autoencoder with 10,240 sparse features over 320-dimensional hidden states. The experiment uses the unnormalized SAE weights released with INTERPLM.

Baselines. We compare sparse features against two controls. First, RAW-HIDDEN ranks the 320 raw ESM-2 layer-4 hidden dimensions using the same Spearman protocol. This checks whether sparse features add value beyond arbitrary dense coordinates. Second, SHUFFLED-SAE shuffles SAE activations across positions within an assay, preserving the marginal activation distribution while breaking residue identity. We use the 95th percentile of shuffled best-feature correlations over 100 repeats as a multiple-comparison-aware control for short sequences and 10,240 tested features. We also report the Spearman correlation between ESM-2 masked-marginal residue sensitivity and DMS sensitivity as a standard zero-shot functional baseline.

Known-motif filtering. We scan each target sequence for ELM and PROSITE motif matches. For top feature-assay pairs, active positions are compared to motif-positive positions on the target sequence. We also evaluate candidate features on the sampled SWISS-PROT proteins by treating motif-positive residues as labels and measuring motif AUROC. A candidate passes the low-alignment heuristic only when it has zero target motif overlap among active positions and near-random SWISS-PROT motif alignment.

Intervention. For selected low-alignment candidates, we ablate the candidate feature’s decoder direction during masked-marginal scoring. At each masked residue position, we run the model with and without the feature-direction ablation at layer 4 and measure the absolute change in mutation log-probability scores. We compare high-activation and low-activation residue groups using the high-minus-low mean absolute score change and a Mann-Whitney test.

Statistics and reproducibility. We use 200 permutations per feature for nominal feature-level permutation tests and Benjamini-Hochberg correction across the 10,240 sparse features. We use paired one-sided Wilcoxon tests across assays for aggregate comparisons against RAW-HIDDEN and SHUFFLED-SAE. Bootstrap 95% confidence intervals summarize medians and median differences. The full run uses random seed 42, CUDA on NVIDIA RTX A6000 GPUs, masked-position batch size 128, and runs in about 20–21 seconds after cached model files are available.

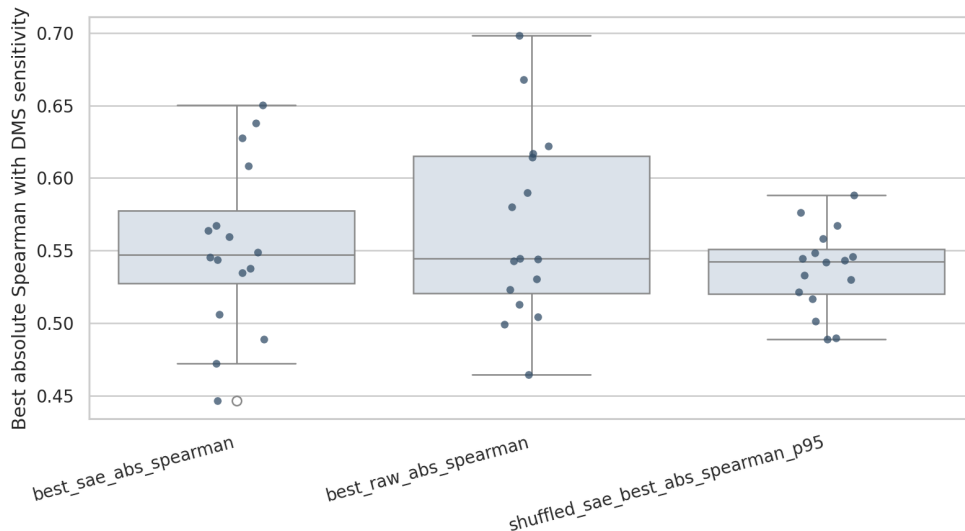


Figure 1: Aggregate baseline comparison across 16 PROTEINGYM assays. Sparse features often correlate with residue-level DMS sensitivity, but the best SAE feature per assay is statistically indistinguishable from the strongest raw hidden dimension and from shuffled sparse-feature controls in this pilot.

A seeded rerun produced byte-identical `assay_summary.csv`, `candidate_features.csv`, and `intervention_results.csv`.

4 Results

Sparse features correlate with DMS sensitivity, but do not beat controls. Figure 1 and table 1 summarize the aggregate comparison. The best SAE feature in each assay reaches a median absolute Spearman correlation of 0.547 with residue-level DMS sensitivity. This is a strong association for short sequences, but it is not unique to the sparse representation: the median best raw hidden-dimension correlation is 0.544, and the median shuffled-feature 95th percentile is 0.543. The SAE best feature exceeds the raw hidden baseline in 8 of 16 assays and exceeds the shuffled 95th percentile in 9 of 16 assays. Paired one-sided Wilcoxon tests do not support an aggregate SAE advantage over raw dimensions ($p = 0.798$) or shuffled controls ($p = 0.217$).

The bootstrap confidence intervals reinforce the same conclusion. The median best SAE correlation has 95% CI 0.526–0.578, while raw hidden dimensions have 95% CI 0.523–0.615 and shuffled controls have 95% CI 0.522–0.552. The median SAE-minus-raw difference is -0.0035 with 95% CI $[-0.052, 0.027]$, and the median SAE-minus-shuffled difference is 0.0089 with 95% CI $[-0.025, 0.055]$. These intervals cross zero, so the aggregate data support feature triage but not a claim that sparse features are reliably better functional markers in this setting.

Low-motif-alignment candidates are rare. We next ask whether high-ranking feature-assay pairs remain after excluding features aligned with implemented motif scans. Figure 2 and table 2 show the three pairs that pass the low target-motif and low SWISS-PROT motif-alignment heuristic. All three have zero target motif overlap among active positions and SWISS-PROT motif AUROC close to 0.5. Their correlations with DMS sensitivity are moderate, with Spearman correlations from -0.389 to -0.484 .

The candidate table also shows why the novelty claim must remain conservative. All three candidates have Benjamini-Hochberg $q = 1.0$ over the full feature set. This is expected under the limited 200-permutation budget and the 10,240 tested sparse features, but it means that nominal permutation p -values from 0.00498 to 0.0199 should be used only for ranking, not confirmatory inference.

One candidate intervention changes masked-marginal scores. We ablate the decoder directions for the selected candidates during ESM-2 masked-marginal scoring. Figure 3 and ta-

Metric	Result
Assays	16
Total single substitutions	12,085
Total residue positions	688
Median best SAE abs. Spearman with DMS	0.547
Median best raw hidden-dim abs. Spearman	0.544
Median shuffled SAE 95th percentile	0.543
SAE > raw assays	8 / 16
SAE > shuffled 95th percentile assays	9 / 16
Wilcoxon SAE > raw p -value	0.798
Wilcoxon SAE > shuffled 95th p -value	0.217
Median ESM-2 score vs. DMS Spearman	0.340
Best SAE median 95% CI	[0.526, 0.578]
Best raw median 95% CI	[0.523, 0.615]
SAE minus raw median 95% CI	[-0.052, 0.027]
SAE minus shuffled median 95% CI	[-0.025, 0.055]

Table 1: Aggregate performance across 16 PROTEINGYM assays. Best absolute correlation is bolded, but paired tests and confidence intervals do not support a reliable aggregate SAE advantage over raw hidden dimensions or shuffled sparse controls.

Assay	Feature	Rank	DMS Spearman	ESM-2 Spearman	Target overlap	SWISS-PROT AUC
TCRG1_MOUSE_Tsuboyama_2023_1E0L	7309	4	-0.484	-0.262	0.000	0.508
YNZC_BACSU_Tsuboyama_2023_2JVD	6419	5	-0.400	0.028	0.000	0.486
TCRG1_MOUSE_Tsuboyama_2023_1E0L	7168	19	-0.389	-0.100	0.000	0.502

Table 2: Low-motif-alignment candidate feature-assay pairs. All three have zero target motif overlap among active positions and near-random SWISS-PROT motif AUROC. Best absolute DMS association is bolded.

ble 3 report the high-minus-low activation effect. Two TCRG1_MOUSE_Tsuboyama_2023_1E0L candidates show negligible or non-significant differences. In contrast, feature 6419 in YNZC_BACSU_Tsuboyama_2023_2JVD changes mutation scores more at high-activation positions than low-activation positions, with an effect of 0.0223 log-prob units and Mann-Whitney $p = 0.00495$.

Feature 6419 is therefore the strongest follow-up candidate in this run. The result is causal with respect to the model computation, because modifying the feature direction changes masked-marginal scores. It is not yet causal with respect to biology. The intervention was selected after screening, the effect is measured in model log-probability units rather than experimental fitness, and the feature does not strongly correlate with ESM-2 residue sensitivity. We treat it as an exploratory candidate for structural or biochemical follow-up.

5 Discussion

What the evidence supports. The strongest supported claim is that FEATURETRIAGE is a useful triage workflow. It connects four pieces of evidence that are usually separated: residue-level sparse activations, DMS sensitivity, known-motif exclusion, and internal model intervention. The

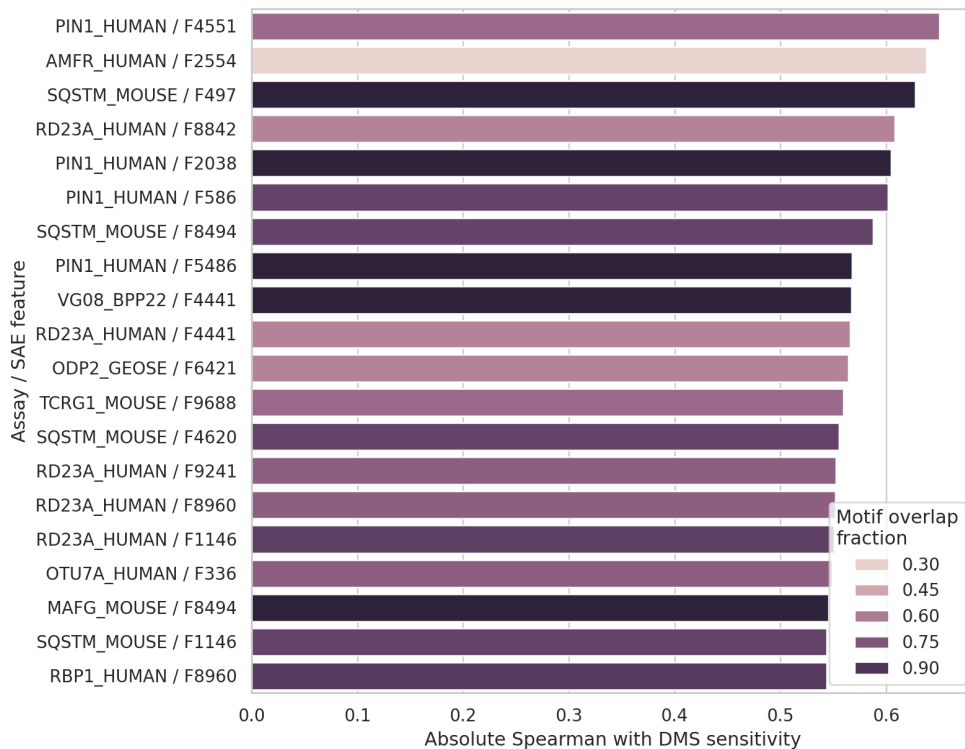


Figure 2: Top low-motif-alignment candidate features. The screen identifies three feature-assay pairs whose active target positions do not overlap the implemented ELM/PROSITE motif scan and whose SWISS-PROT motif AUROC is near random. These are candidate hypotheses, not discoveries, because feature-level FDR correction does not support significance.

Assay	Feature	Mutations	Positions	High-low effect	Mann-Whitney <i>p</i>
TCRG1_MOUSE.Tsuboyama_2023_1E0L	7309	621	35	0.00233	0.254
YNZC_BACSU.Tsuboyama_2023_2JVD	6419	714	39	0.02228	0.00495
TCRG1_MOUSE.Tsuboyama_2023_1E0L	7168	621	35	0.00013	0.498

Table 3: Decoder-direction ablations for low-motif-alignment candidates. The effect is the mean absolute masked-score change at high-activation positions minus the corresponding change at low-activation positions. Best effect and smallest *p*-value are bolded.

workflow surfaces candidates that are plausible enough for follow-up while making the uncertainty visible.

What the evidence does not support. The experiment does not establish a new biological mechanism. The aggregate association results are close across SAE, RAW-HIDDEN, and SHUFFLED-SAE controls. This matters because a sparse feature can be interpretable without being a distinct biological abstraction. It also matters because 10,240 sparse features tested on sequences of length 37–52 create many chances to find high correlations. In this setting, raw hidden dimensions and shuffled activations are not weak baselines; they are necessary checks against overclaiming.

Interpreting feature 6419. Feature 6419 is interesting because it passes the implemented motif filters and its ablation changes ESM-2 masked-marginal scores more at high-activation positions. This suggests that the feature participates in the model’s mutation-score computation for the YNZC_BACSU target. The next question is biological coherence. High-activation positions should be tested for structural proximity, solvent exposure, conservation, physicochemical patterning, and consistency across homologs. Without those checks, the feature remains a model-internal hypothesis rather than an inferred mechanism.

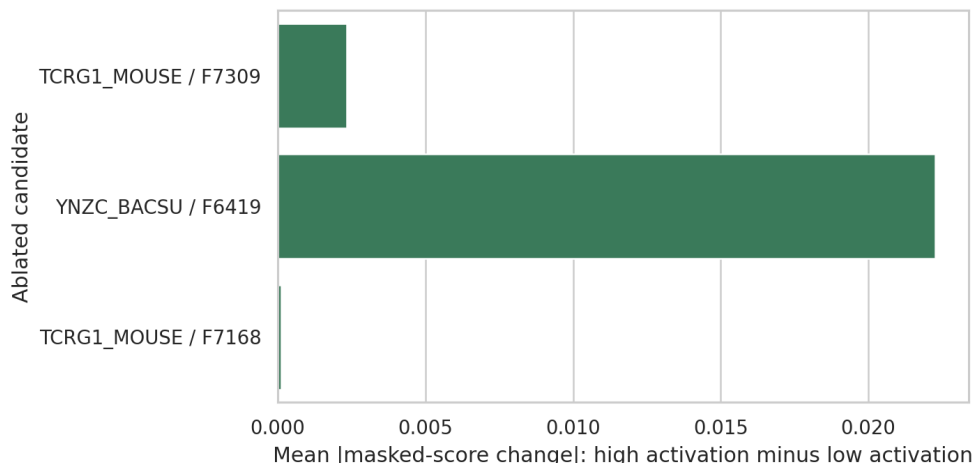


Figure 3: Internal intervention effects for the three low-motif-alignment candidates. Ablating feature 6419 in the YNZC_BACSU assay produces the clearest high-minus-low activation effect on masked-marginal mutation scores, while the two TCRG1_MOUSE candidates do not show comparable separation.

Limitations. The pilot uses ESM-2-8M for tractability, while larger ESM-2 models may contain cleaner or different sparse features. The assay panel contains 16 short proteins, mostly from Tsuboyama-style stability assays, so it is not broad biological coverage. The PROSITE converter covers common pattern syntax but not all profiles or rules. The ELM/PROSITE scans are permissive and incomplete, so “low motif alignment” means low alignment under the implemented scan, not absence of known biology. The feature-level permutation budget is small relative to 10,240 tested features, so candidate q -values do not survive FDR correction. Finally, internal ablation effects are measured in masked-marginal log-probability units, not experimental fitness.

Broader implications. A cautious workflow is valuable because PLM-derived biological claims can otherwise move faster than the evidence. Feature triage can help prioritize structural analysis and mutagenesis, but it should not replace annotation curation or experimental validation. The same caution also reduces misuse: model-internal signals should not be treated as sufficient evidence for functional annotation, disease interpretation, or protein-design decisions without independent checks.

6 Conclusion

We tested whether sparse INTERPLM features inside ESM-2-8M identify functionally important protein residues that are not explained by implemented ELM/PROSITE motif scans. The answer from this pilot is mixed and conservative. Sparse features often correlate with PROTEINGYM residue sensitivity, but they do not significantly outperform raw hidden dimensions or shuffled sparse controls across 16 short assays. The workflow identifies three low-motif-alignment candidates and one exploratory intervention signal, feature 6419 in YNZC_BACSU_Tsuboyama_2023_2JVD, where ablation changes masked-marginal scores preferentially at high-activation positions.

The key takeaway is that PLM internals can generate useful biological hypotheses, but novelty claims require baseline controls and causal checks. Future work should repeat the protocol on larger ESM-2 models and more PROTEINGYM assays, add structure-aware and evolutionary validation, use more complete motif and UniProt annotations, increase permutation counts or use max- T inference, and design targeted variants that test whether feature-preserving substitutions behave differently from feature-disrupting substitutions.

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